A Computational Heuristic and Visualization Framework for the Riemann Zeta Function and its Inverse

[Sajid Hussain]
Independent Researcher Gilgit-Baltistan Pakistan

March 28, 2026

Abstract

This paper presents a computational heuristic and visualization framework mapping the relationship between the Riemann zeta function, $\zeta(s)$, and its multi-valued inverse, $\zeta^{-1}(w)$. By constructing the bounding correspondence $2\cosh(sw) = e^{\zeta(s)} + e^{\zeta^{-1}(w)}$, we establish a numerically stable wrapper for evaluating these functions across the complex plane. Utilizing 50-decimal-place precision, we demonstrate an asymptotic proportional collapse of the relative error as $s \to \infty$ on the real line. Furthermore, evaluating this correspondence at the non-trivial zeros mathematically simplifies the system to $1 - e^{\rho}$. Assuming the Riemann Hypothesis, this maps the infinite critical line onto a closed, finite circular boundary of radius $\sqrt{e}$. We discuss the utility of this exponential compaction as a geometric anomaly detector and an educational visualization tool for complex analysis.

Keywords: Riemann Zeta Function, Experimental Mathematics, Visualization, Computational Heuristics, Dimensional Compaction.

1 Introduction

The multi-valued nature of the inverse Riemann zeta function, $\zeta^{-1}(w)$, presents computational challenges, as standard numerical evaluations often suffer from branch-jumping. This paper introduces a computational heuristic designed to geometrically bound and stabilize the evaluation of these functions. By utilizing a log-sum-exp structural format, we create a stable, hyperbolic bounding surface. While this study does not attempt to prove the Riemann Hypothesis, it provides a novel geometric mapping that compacts the infinite critical strip into a finite phase space, serving as both a stable algorithmic wrapper and a visual discriminator for computational number theory.

2 The Core Heuristic and Real Asymptotics

We propose the following hyperbolic bounding surface relating s and w:

$$2\cosh(sw) = e^{\zeta(s)} + e^{\zeta^{-1}(w)}$$

To empirically verify this structural correspondence, we restrict our evaluation to the principal branch curve where $w = \zeta(s)$, yielding the absolute error function:

$$F(s) = 2\cosh(s\zeta(s)) - e^{\zeta(s)} - e^{s}$$

Numerical verification using 50-decimal-place precision demonstrates that while the absolute error diverges exponentially as $s \to \infty$ for $s \in \mathbb{R}$, the relative error undergoes a strict proportional collapse:

$$\lim_{s\to\infty} \left| \frac{2\cosh(s\zeta(s)) - e^{\zeta(s)}}{e^{s}} \right| - 1 = 0$$

This asymptotic freeze indicates that the bounding surface becomes proportionally exact at positive infinity.

3 Complex Tracking via Homotopy Continuation

Extending this correspondence to $\mathbb{C}^2$ requires navigating the infinite branch cuts of $\zeta^{-1}(w)$. Because the exponential function is periodic ($e^{z+2\pi i} = e^{z}$), placing the inverse zeta function inside an exponent acts as a topological stabilizer. Computational models utilizing Homotopy Continuation confirmed that this bounding surface maintains its structural integrity under complex rotation, providing a reliable numerical guardrail against branch-jumping errors.

4 Geometric Projection of the Critical Line

A visually profound behavior of this heuristic emerges when evaluated at the non-trivial zeros of the zeta function, where $s = \rho$ and $w = \zeta(\rho) = 0$.

Plugging these coordinates into the error function yields:

$$F(\rho) = 2\cosh(0) - e^{0} - e^{\rho} = 1 - e^{\rho}$$

This simplification relies on a fundamental property of complex exponentials. If we express any complex number as $s = x + it$, its exponential magnitude is dictated entirely by its real part: $|e^{x+it}| = e^x$.

Assuming the Riemann Hypothesis ($\Re(\rho) = 1/2$), our correspondence naturally maps every non-trivial zero onto a closed circle centered at $z = 1$ with a strict radius:

$$R = |e^{1/2+it}| = e^{1/2} = \sqrt{e} \approx 1.648721$$

Because the sequence of imaginary heights t_n is aperiodic, the mapped points do not overlap; rather, they continuously wrap around the perimeter, asymptotically generating a dense ring.

5 Discussion: Utility of the Framework

5.1 A Geometric Filter for Anomaly Detection

The primary advantage of this exponential mapping is "dimensional compaction." Searching for a counterexample to the Riemann Hypothesis traditionally involves tracking deviations along an infinitely tall, open-ended critical line. By passing the zeros through the $1 - e^{\rho}$ filter, the imaginary height t is entirely converted into a rotational angle, while the real part strictly governs the radius. Therefore, if a "rogue zero" exists off the critical line (e.g., $\Re(s) = 0.8$), it will immediately break the $\sqrt{e}$ radial boundary and be geometrically expelled or trapped by the circle. This provides an elegant, bounded algorithm for visual anomaly detection.

5.2 Pedagogical and Computational Stability

Mathematically, tracking the relationship between a complex function and its multi-valued inverse is notoriously difficult to visualize. By wrapping the inverse zeta function in a hyperbolic cosine correspondence, we stitch the chaotic sheets of the Riemann surface into a manageable metric. This acts as a robust educational tool for demonstrating asymptotic behaviors, branch cut stabilization, and complex mapping to students and researchers in experimental mathematics.

6 Conclusion

This experimental study successfully demonstrates a numerically stable, asymptotic bounding heuristic for the Riemann zeta function and its inverse. By reframing the evaluation of these functions through the correspondence $2\cosh(sw) = e^{\zeta(s)} + e^{\zeta^{-1}(w)}$, we provide a highly bounded geometric discriminator. While rooted in standard exponential mechanics, this dimensional compaction offers computational mathematicians a stable, visually intuitive algorithm for tracking roots and visualizing the complex phase space of the critical strip.

References

[1] Riemann, B. (1859). *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse.* Monatsberichte der Berliner Akademie.

[2] Titchmarsh, E. C. (1986). *The Theory of the Riemann Zeta-function* (2nd ed.). Oxford University Press.

[3] Borwein, J., Bailey, D. (2004). *Mathematics by Experiment: Plausible Reasoning in the 21st Century.* A K Peters/CRC Press.